

TITAN A16 - REQUEST FOR QUOTE (RFQ)
Precision Machining Services - Research Prototype

Issued by: Matthew James Campbell
Date: January 22, 2026
RFQ Number: TITAN-A16-RFQ-001
Quantity: 2 units (reproducibility validation)
Delivery Required: [YOUR DEADLINE]

1. PROJECT OVERVIEW

Part Name: TITAN A16 Resonant Test Article
Application: Experimental resonance research device
Material: Monolithic Ta-Ti-Au-C alloy billet (custom composition)
Critical Features: Precision bore, triple-helix geometry, 12-node top lattice

This is a research prototype requiring exceptional dimensional control and surface finish.

2. MATERIAL REQUIREMENTS

2.1 Base Material Supply

OPTION A (Preferred): Bidder sources custom billet

- Composition: Ta (primary), Ti (secondary), Au (interfacial), C (trace)
- Minimum purity: ≥99.95% per element
- Total impurity: <0.001%
- Grain size: ≤15 µm preferred
- Billet dimensions: Ø32mm × 12mm minimum (allow machining stock)
- Processing: Vacuum Induction Melting (VIM) required

OPTION B: Client supplies pre-cast billet

- Bidder machines from supplied stock
- Material certification (composition, purity) provided by client

Material certification required: Mill test report, composition analysis (ICP-MS or equivalent)

3. PART SPECIFICATIONS

3.1 Overall Dimensions

Feature	Specification	Tolerance
	----- ----- -----	
Outer Diameter (OD)	Ø30.00 mm	+0.03 / -0.00
Total Height	10.00 mm	±0.05 mm
Central Bore Diameter	Ø3.03 mm	±0.03 mm
Bore Concentricity to OD	—	±0.02 mm

3.2 Axial Layer Stack

Five internal material layers (L1–L5), each 2.00mm ±0.03mm thick:

- L1, L3, L5: Titanium-dominant zones
- L2, L4: Gold-dominant interfacial zones

(Note: This is a composition gradient within the monolithic billet, NOT assembled laminate layers)

3.3 Top Lattice Features

- **12 node pockets** arranged in hexagram pattern
- Pocket diameter: Ø2.00mm ±0.03mm
- Pocket depth: 1.00mm ±0.03mm
- Radial position: 5.00mm from bore centerline
- Angular spacing: 30° increments (0°, 30°, 60°... 330°)
- Coordinate table provided in Section 9

3.4 Triple Helix Geometry

- Three interwoven helical threads on outer surface
- Helix radius: 12.5mm from centerline
- Pitch: 3.33mm per revolution (3 complete turns over 10mm height)
- Angular offset: 120° between starts
- Thread depth: 0.5mm nominal
- *Detailed G-code provided if selected*

3.5 Surface Finish

- All functional surfaces: **Ra 0.8 – 1.6 µm**
- No directional scoring permitted on top/bottom faces
- Bore surface: Ra ≤1.0 µm

4. MANUFACTURING REQUIREMENTS

4.1 Machine Capabilities Required

-  5-axis CNC machining center
-  Deep-hole drilling capability (Ø3mm × 10mm depth)

-  Reaming to $\pm 0.03\text{mm}$ on small bores
-  High-speed finishing (8,000-12,000 RPM)
-  PCD (polycrystalline diamond) tooling for final passes

4.2 Thermal Processing

Post-machining stress relief required:

- Temperature: $540^{\circ}\text{C} \pm 10^{\circ}\text{C}$
- Duration: 2 hours
- Atmosphere: Argon or vacuum ($<10^{-4}$ Torr)
- Cooling: Controlled furnace cool to $<100^{\circ}\text{C}$ before removal

4.3 Optional Enhancement

****Ultrasonic Conditioning (if equipment available):****

- Frequency: $360\text{ Hz} \pm 5\text{ Hz}$
- Duration: 15 minutes
- Purpose: Resonance stabilization
- *Not required, but preferred*

5. INSPECTION & QUALITY CONTROL

5.1 Dimensional Inspection

All critical dimensions verified via ****Coordinate Measuring Machine (CMM)****:

- OD, height, bore diameter
- Bore concentricity to OD datum
- Node pocket positions (X/Y coordinates)
- Node pocket depths

****First Article Inspection (FAI) report required**** for Unit 1

5.2 Surface Finish Verification

- Profilometer measurement on 3 locations per functional surface
- Data sheet provided with part

5.3 Material Verification

- Grain size analysis (SEM or EBSD preferred)
- Composition spot-check via EDS/XRF (if in-house capability exists)

5.4 Resonance Test (Optional)

If shop has vibration analysis equipment:

- Impulse excitation test
- Record frequency response 100-1000 Hz
- Target resonance peak: $360\text{ Hz} \pm 10\text{ Hz}$

6. DELIVERABLES

Per Unit:

- ☒ Machined TITAN A16 part
- ☒ CMM inspection report (full dimensional data)
- ☒ Surface finish data sheet
- ☒ Material certification (composition, purity)
- ☒ Grain size analysis report
- ☒ Photographic documentation (6 views minimum)

For Lot (2 units):

- ☒ First Article Inspection (FAI) package
- ☒ Process traveler / work order record
- ☒ Tool list and cutting parameters used

7. PACKAGING & SHIPPING

- Individual polyethylene bags (non-reactive)
- Rigid foam inserts in shipping box
- Anti-static precautions (if applicable)
- Shipped via trackable carrier with insurance

****Handle with non-marring gloves only. No direct skin contact.****

8. BIDDING INSTRUCTIONS

8.1 Quote Submission Requirements

Please provide:

1. ****Unit price**** (per part, for qty 2)
2. ****Tooling/setup costs**** (one-time, if applicable)
3. ****Material sourcing approach**** (Option A or B from Section 2)
4. ****Lead time**** (from material receipt to shipment)
5. ****Payment terms****
6. ****Shop capabilities statement:****
 - Machine models available
 - CMM/inspection equipment

- Thermal processing equipment
- Previous exotic alloy experience

7. ****References**** (similar precision work, preferably Ta/Ti alloys)

8.2 Selection Criteria

- Dimensional capability and tolerance experience (40%)
- Material handling experience (25%)
- Inspection/QC infrastructure (20%)
- Price (10%)
- Lead time (5%)

8.3 Submission Deadline

****[INSERT DEADLINE - suggest 14 days from issue]****

Submit to:

****Matthew James Campbell****

****[YOUR EMAIL]****

****[YOUR PHONE]****

9. APPENDIX A: TOP LATTICE NODE COORDINATES

Reference datum: Bore centerline (X0 Y0), top face (Z0)

Node	Angle (°)	X (mm)	Y (mm)	Material Zone	Depth (mm)
-----	-----	-----	-----	-----	-----
1	0	+5.000	0.000	Ti	1.00
2	30	+4.330	+2.500	Au	1.00
3	60	+2.500	+4.330	Ti	1.00
4	90	0.000	+5.000	Au	1.00
5	120	-2.500	+4.330	Ti	1.00
6	150	-4.330	+2.500	Au	1.00
7	180	-5.000	0.000	Ti	1.00
8	210	-4.330	-2.500	Au	1.00
9	240	-2.500	-4.330	Ti	1.00
10	270	0.000	-5.000	Au	1.00
11	300	+2.500	-4.330	Ti	1.00
12	330	+4.330	-2.500	Au	1.00

10. APPENDIX B: MATERIAL PROPERTIES (REFERENCE)

For FEA/thermal analysis planning:

Property	Tantalum	Titanium	Gold
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Density (g/cm ³)	16.65	4.51	19.32
Young's Modulus (GPa)	186	116	79
Poisson's Ratio	0.34	0.32	0.42
Thermal Expansion (10 ⁻⁶ /K)	6.3	8.6	14.2
Melting Point (°C)	3,017	1,668	1,064

11. QUESTIONS & CLARIFICATIONS

****Technical contact:****

Matthew James Campbell

****[YOUR CONTACT INFO]****

****Pre-bid conference:**** Available upon request (video call)

12. TERMS & CONDITIONS

- Client reserves right to reject any/all bids
- Material certification required before machining begins
- Client may visit shop for FAI witness (with 48hr notice)
- Payment upon delivery and acceptance of dimensional inspection
- Scrap material remains property of client (return required)

****END OF RFQ DOCUMENT****

This is a research project. We value precision, communication, and documentation quality over speed.